PROJECT PHASE 2

SMART TRANSFORMER MONITORING AND FAULT DETECTION USING IoT SENSORS

GROUP 10

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INTRODUCTION

- Transformers are essential for power distribution in industries and society.
- Normally, manual monitoring of transformers is done and it has several limitations and may fail to detect faults in time.
- To overcome these a smart transformer monitoring and fault detection system using IoT sensors is proposed.

PROJECT OBJECTIVES

- Monitor critical parameters of the transformer in real time.
- Detect abnormalities like overheating, oil level drop, moisture in transformer oil ,etc..
- Enable remote monitoring using IoT and web-based interface.
- Reduce manual inspection and maintenance efforts.

LITERATURE REVIEW

Sl. No	Author Names, year	Paper title	Journal name (SCI/Scopus)	Contents related to work
1.	V. Ravindran, R. Ponraj, C. Krishnakumar, S. Ragunathan, V. Ramkumar and K. Swaminathan, 2021	IoT-Based Smart Transformer Monitoring System with Raspberry Pi	*Innovations in Power and Advanced Computing Technologies (i-PACT)*	• Measures parameters like voltage, current, oil level, temperature. • Uses ADC,sensors,web interface for real-time monitoring.
2.	Z. Talib, Z. Faris and L. Ali Abdul Rahaim, 2024	IoT Monitoring System-based enabling Intelligent power Transformer using Raspberry Pi Technology	*International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA)*	• Raspberry Pi based embedded system to monitor transformer parameters
3.	W. K. A. Hasan, A. Alraddad, A. Ashour, Y. Ran, M. A. Alkelsh and R. A. M. Ajele, 2019	Design and Implementation Smart Transformer based on IoT	*International Conference on Computing, Electronics & Communications Engineering (iCCECE)*	• Proposes an IoT-based real-time transformer monitoring system using sensors.
4.	R. R. Pawar, P. A. Wagh and S. B. Deosarkar, 2017	Distribution transformer monitoring system using Internet of Things (IoT)	*International Conference on Computational Intelligence in Data Science(ICCIDS)*	• Provides real-time alerts via SMS and web interface during abnormal conditions.

TRANSFOMER FAULTS AND LIMITS

Our system is designed to monitor key parameters and detect faults by comparing sensor readings to predefined limits.

Current Sensor

- Measures load current ,detects overcurrent and short-circuits.
- 10-20% above rated current- mild overload
- Greater than 25-30% above rated current – current fault

Voltage Sensor

- Monitors output voltage, detects under/over-voltage.
- Greater than 15% of rated voltage- fault condition

TRANSFOMER FAULTS AND LIMITS

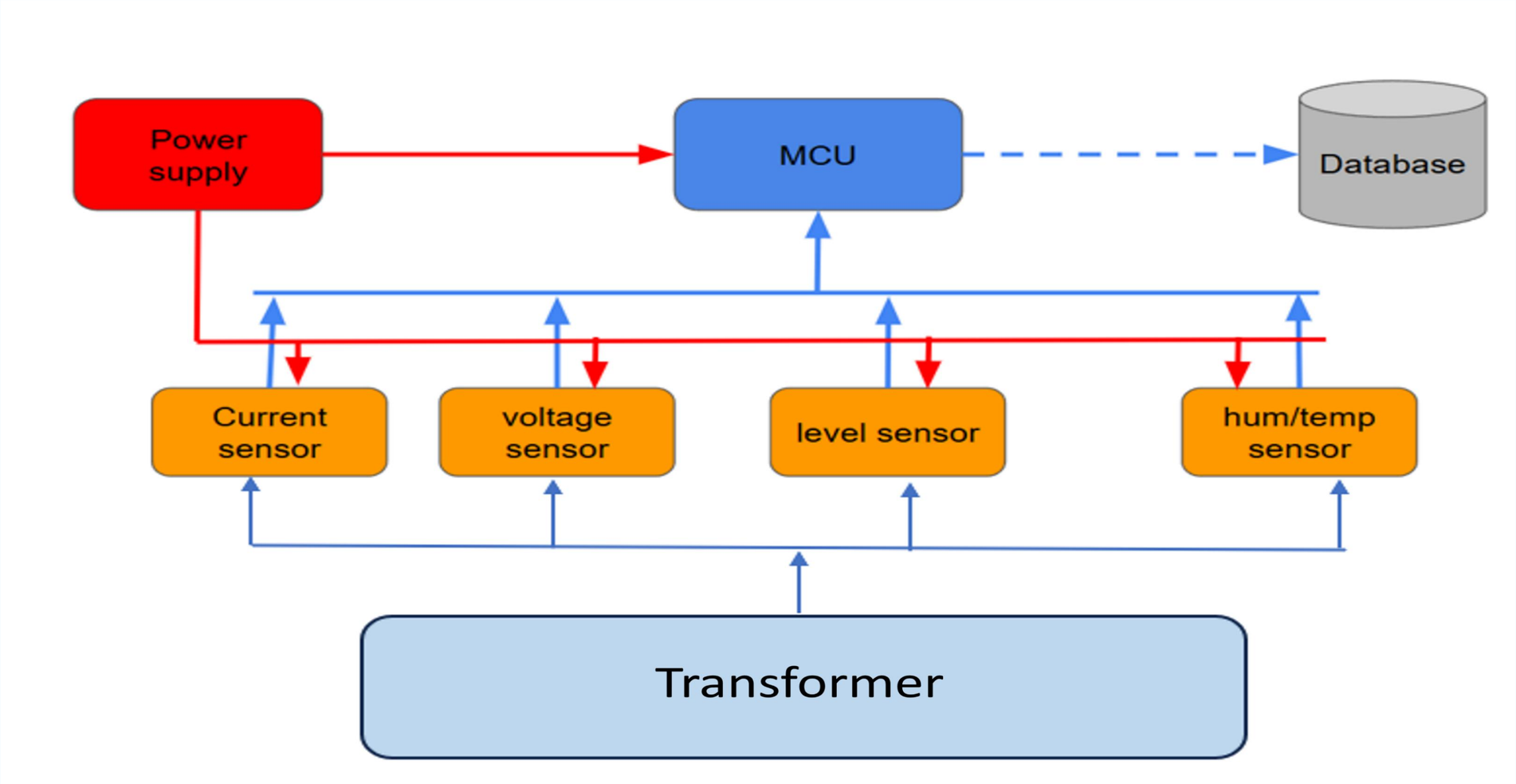
Oil Level Sensor

- Detects oil leakage or oil level ,prevents damage due to poor cooling/insulation.

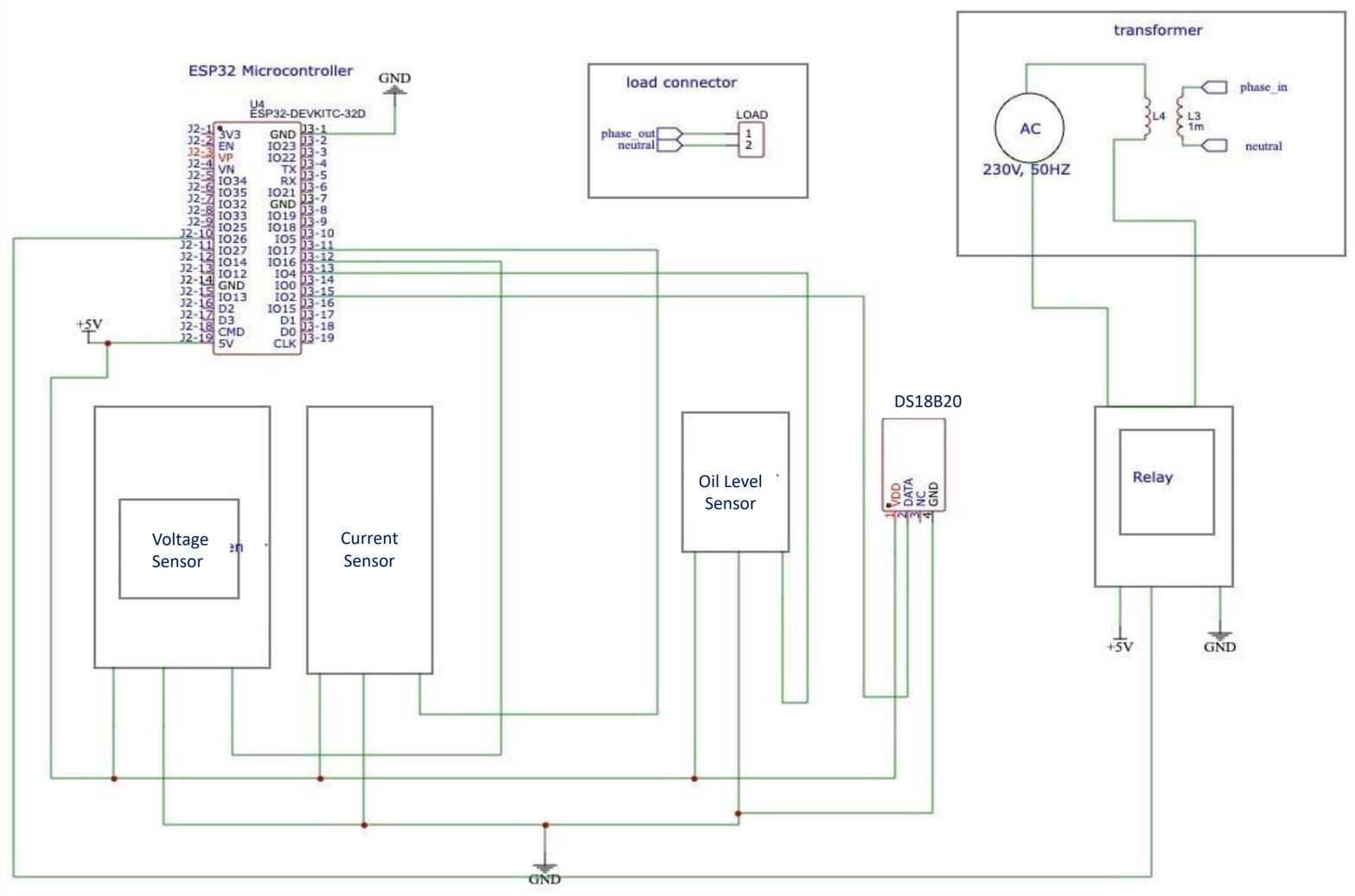
Temperature Sensor

- Monitors oil temperature, prevents overheating and insulation damage.
- Temperature > 80 degree Celsius – oil degradation

BLOCK DIAGRAM



CIRCUIT DIAGRAM



COMPONENTS

- Voltage Sensor
- Current Sensor
- Temperature Sensor
- Oil Level Sensor
- Relay Module
- ESP32

Updated Sensors



- Voltage Sensor
- Input voltage-0-25V



- Current Sensor – ACS712
- Input Current Range: 0–5A AC

Temperature sensor-DS18B20

- DS18B20 is a temperature sensor used for accurate temperature measurement.
- Measures temperature in the range of −55°C to +125°C with ±0.5°C accuracy.



Sharp Distance Measuring Sensor

- Infrared-based distance measurement sensor used for proximity and object detection.
- Output type: Analog.



Relay Module

- 5V Single Channel Relay Module.
- Provides electrical isolation between control circuit and load.

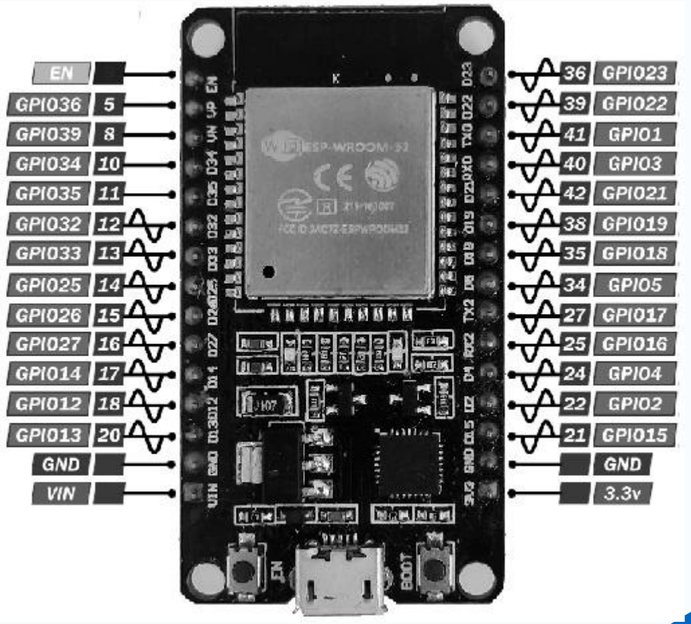


ESP-32

- Purpose: Wi-Fi + Bluetooth microcontroller used for IoT, control and data processing.

Technical Details:

- Processor: Dual-core Tensilica LX6 (up to 240 MHz)
- RAM: 520 KB SRAM
- Flash Memory: 4 MB (typical)
- ADC Channels: 18 (12-bit resolution)



FAULT MANAGEMENT

OVER/UNDER-VOLTAGE FAULT

- Isolate transformer until supply stabilizes.
- Reduce the load.

CURRENT FAULTS

- Disconnect load immediately.
- Inspect cables for damaged connection and check for winding insulation failure.

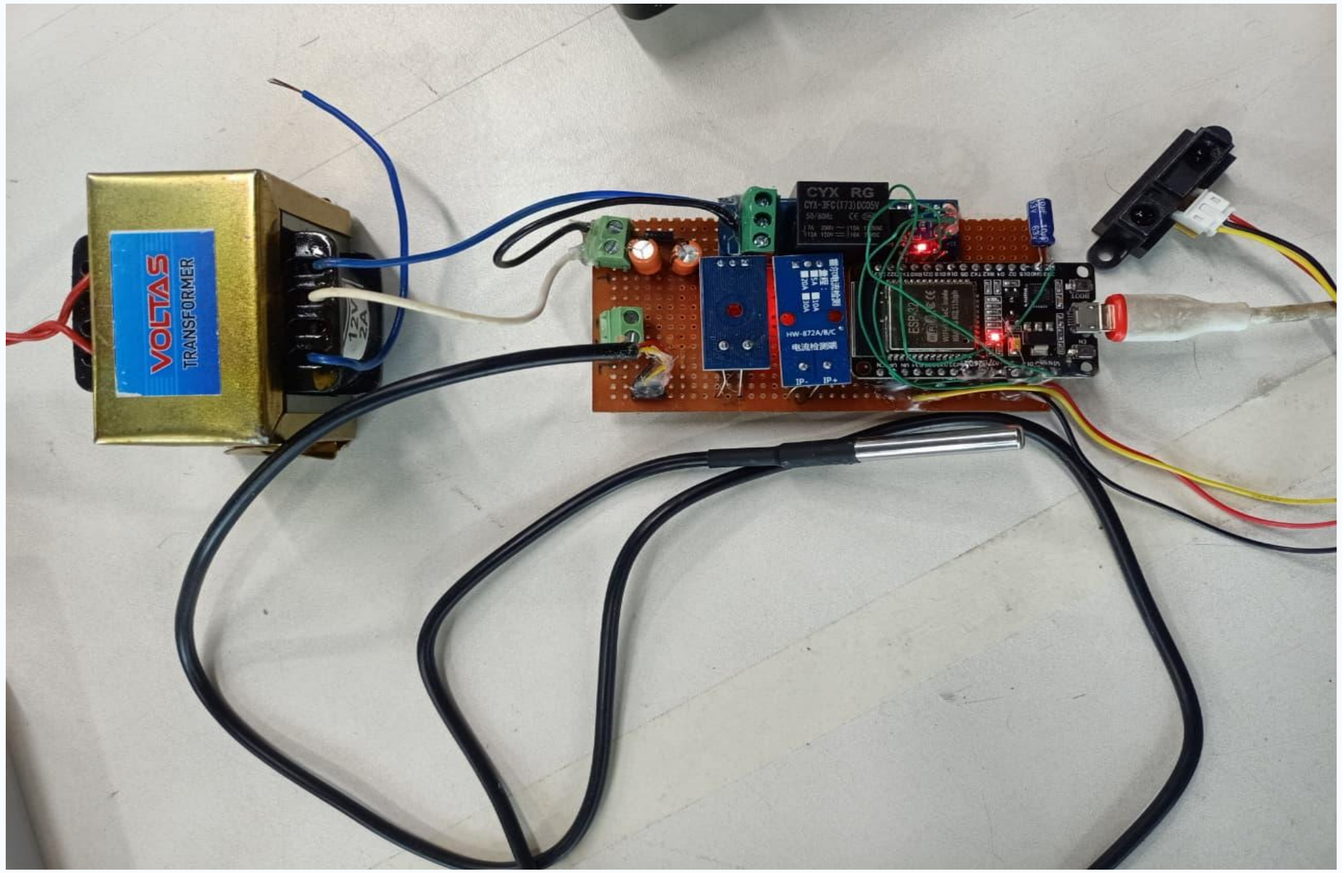
OIL LEVEL FAULT

Identify leak point and seal it.

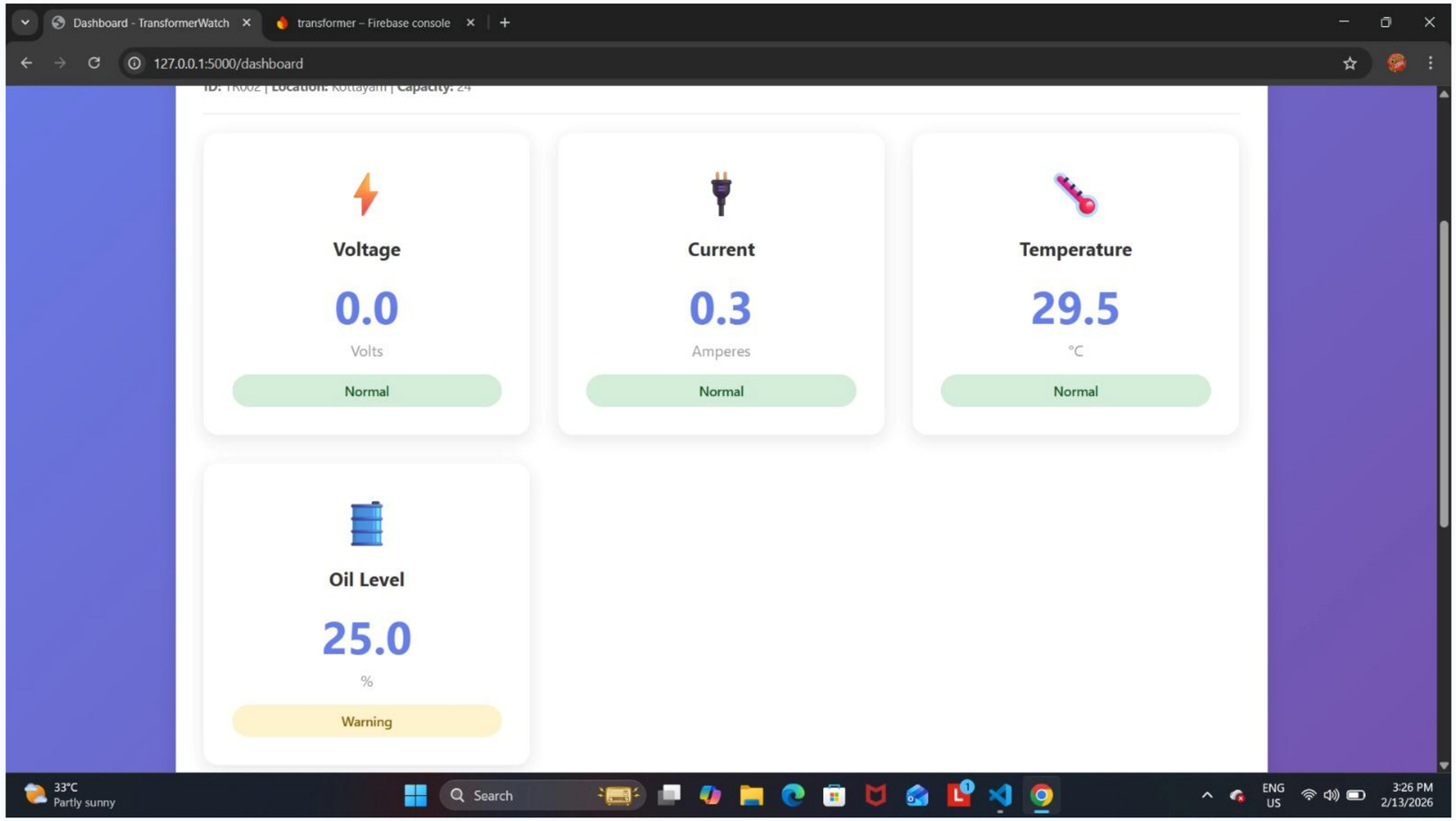
HIGH TEMPERATURE

- Immediate load reduction.
- Inspect cooling fans, radiators, pumps.

HARDWARE

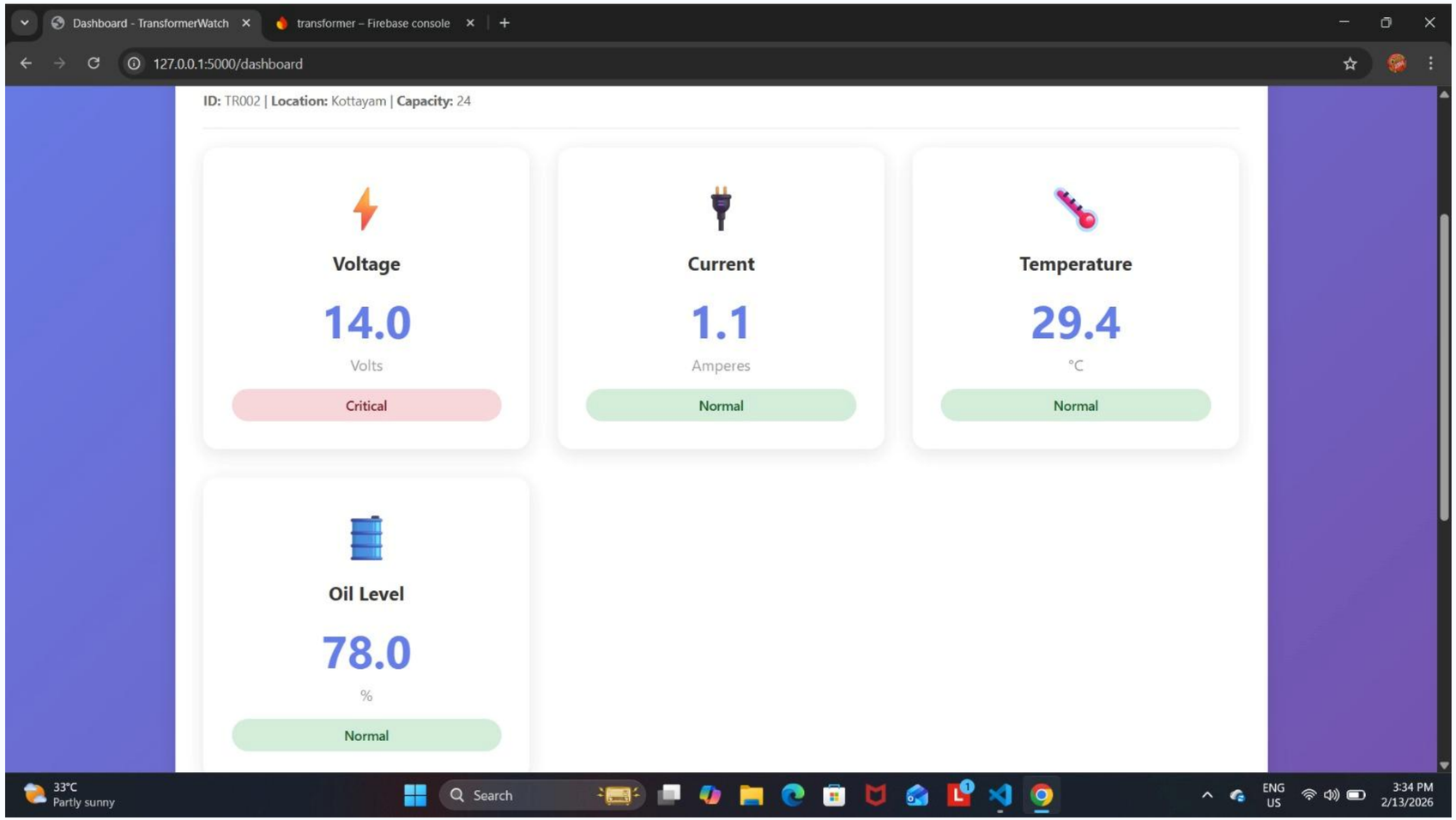


RESULT


Dashboard - TransformerWatch
transformer – Firebase console
127.0.0.1:5000/dashboard
Voltage
0.0
Volts
Normal
Current
0.3
Amperes
Normal
Temperature
29.5
°C
Normal
Oil Level
25.0
%
Warning
33°C
Partly sunny
Search
ENG
US
3:26 PM
2/13/2026



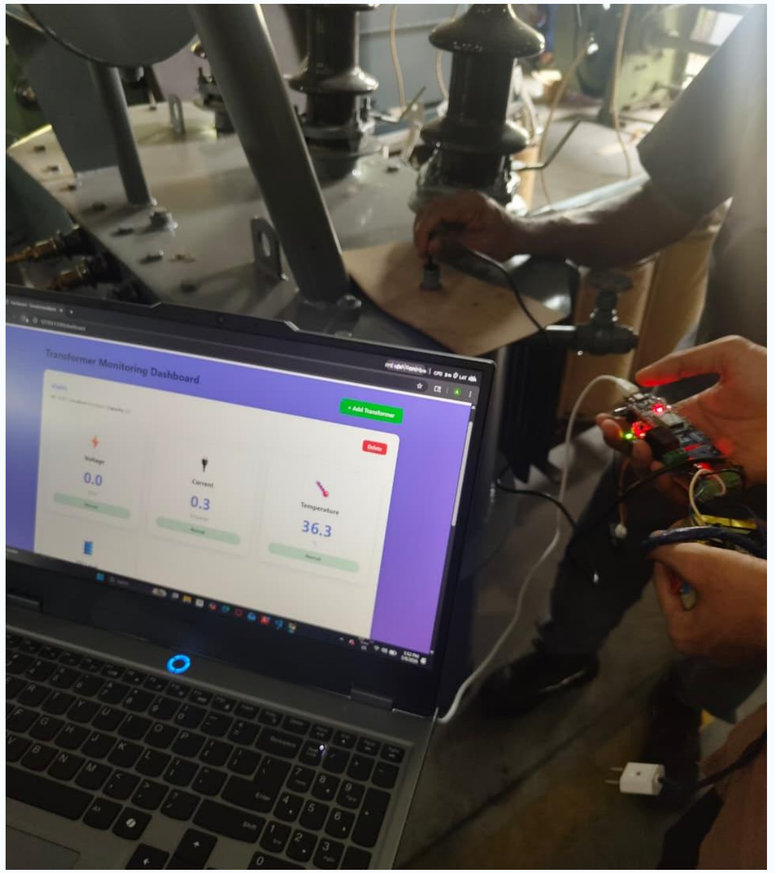
RESULT


Dashboard - TransformerWatch
transformer – Firebase console
127.0.0.1:5000/dashboard
ID: TR002 | Location: Kottayam | Capacity: 24
Voltage
14.0
Volts
Critical
Current
1.1
Amperes
Normal
Temperature
29.4
°C
Normal
Oil Level
78.0
%
Normal
33°C
Partly sunny
Search
ENG
US
3:34 PM
2/13/2026



HARDWARE TESTING AT KEL MAMALA





CONCLUSION

- Successfully developed the system prototype measuring the critical parameters and its monitoring interface.
- The system provides real-time monitoring of transformer parameters such as voltage, current, temperature, and oil level.
- With IoT integration, users can remotely access data and receive alerts in case of abnormalities.
- This enables faster fault detection, prevents major failures, and improves transformer efficiency, safety, and reliability.

REFERENCES

[1] V. Ravindran, R. Ponraj, C. Krishnakumar, S. Ragunathan, V. Ramkumar and K. Swaminathan, "IoT-Based Smart Transformer Monitoring System with Raspberry Pi," *2021 Innovations in Power and Advanced Computing Technologies (i-PACT)*, Kuala Lumpur, Malaysia, 2021

[2] Z. Talib, Z. Faris and L. Ali Abdul Rahaim, "IoT Monitoring System -based enabling Intelligent power Transformer using Raspberry Pi Technology," *2024 International Congress on Human-Computer Interaction, Optimization and Robotic Applications (HORA)*, Istanbul, Turkiye, 2024

[3] K. M. Nathgosavi, S. D. Nangare, S. P. Pol, A. S. Pawar, A. S. Gherade and S. S. Kumbhar, "Real-Time Monitoring of Transformer Parameters Using IoT," *2025 4th OPJU International Technology Conference (OTCON) on Smart Computing for Innovation and Advancement in Industry 5.0*, Raigarh, India, 2025

[4] W. K. A. Hasan, A. Alraddad, A. Ashour, Y. Ran, M. A. Alkelsh and R. A. M. Ajele, "Design and Implementation Smart Transformer based on IoT," *2019 International Conference on Computing, Electronics & Communications Engineering (iCCECE)*, London, UK, 2019

[5] D. Srivastava and M. M. Tripathi, "Transformer Health Monitoring System Using Internet of Things," 2018 2nd IEEE International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES), Delhi, India, 2018

[6] R. R. Pawar, P. A. Wagh and S. B. Deosarkar, "Distribution transformer monitoring system using Internet of Things (IoT)," 2017 International Conference on Computational Intelligence in Data Science(ICCIDS), Chennai, India, 2017

THANK YOU